BSA / BUSINESS STRATEGY & ANALYTICS CLUB

NEXAURA

PROMPT BATTLE

PROBLEM STATEMENTS

BUILD BOLDLY. VERIFY INTELLIGENTLY.
EIGHT CHALLENGES. THREE DIFFICULTY LEVELS. SIX HOURS TO BUILD.

THINK SMART / BUILD FAST / PROVE IT WORKS

31 OCT
6-HOUR HACKATHON

AI × IDEAS × EXECUTION

2–4
MEMBERS PER TEAM

CHOOSE YOUR CHALLENGE

7 distinct problem statements • 3 difficulty levels • One clear challenge per row

PS 01	CropSense	Smart Crop Recommendation	EASY
PS 02	CampusConnect	College Notice Board	EASY
PS 03	CampusLink	AI-Powered Placement & Recruitment Management Platform	MEDIUM
PS 04	ContradictIQ	Financial Intelligence & Evidence Investigation Platform	MEDIUM
PS 05	Strategy Arena	Interactive Business Strategy Simulation & Decision Platform	MEDIUM
PS 06	ConsultOS	Autonomous Business Investigation & Workflow Platform	HARD
PS 07	SignalForge	Quantitative Research & Experiment Management Terminal	HARD

PS 01

CropSense

EASY

Smart Crop Recommendation

Farmers need to select crops that are suitable for their soil and local weather conditions. Choosing an unsuitable crop can affect agricultural productivity. Basic soil and climate information can help identify suitable crop options.

THE CHALLENGE

Build a beginner-friendly Machine Learning application that recommends a suitable crop based on soil nutrients and environmental conditions using a classification model trained on an existing dataset.

BASIC FUNCTIONALITIES

- Load the provided crop recommendation dataset.
- Explore soil and weather parameters such as Nitrogen, Phosphorus, Potassium, temperature, humidity, pH and rainfall.
- Clean and prepare the dataset for model training.
- Train a basic classification model to predict the suitable crop.
- Allow users to enter soil and weather values and get a crop recommendation.
- Display the predicted crop in a simple interface.

ADVANCED FUNCTIONALITIES

- Compare Decision Tree, K-Nearest Neighbors and Random Forest models.
- Display model performance using accuracy and a confusion matrix.
- Visualize the distribution of different crops in the dataset.
- Show which input parameters influence predictions the most.

Dataset Link - https://www.kaggle.com/datasets/atharvaingle/crop-recommendation-dataset

BROWNIE POINTS

- **Display the top three recommended crops with their prediction probabilities.**

PS 02

CampusConnect

EASY

College Notice Board

Students often miss important college announcements, event updates, and notices because information is scattered across different WhatsApp groups and communication channels.

THE CHALLENGE

Build a simple college notice board website where students can view announcements, while an admin can add, edit, and delete notices from a centralized platform.

BASIC FUNCTIONALITIES

- Display all college notices on a simple homepage.
- Allow an admin to add new notices with a title, description, and date.
- Allow an admin to edit and delete existing notices.
- Organize notices into categories such as Events, Academics, and General.
- Provide a search option to find notices by title or keyword.

ADVANCED FUNCTIONALITIES

- Add category-based filtering.
- Highlight important or recent announcements.
- Make the website responsive for mobile and desktop screens.
- Store notices in a database instead of static data.

BROWNIE POINTS

- Add an admin login system so only authorized users can manage college notices.

PS 03

CampusLink

MEDIUM

AI-Powered Placement & Recruitment Management Platform

College placement teams manage student profiles, resumes, job descriptions, eligibility requirements and interviews using spreadsheets and separate communication channels. Students may miss suitable opportunities, while recruiters and placement coordinators spend considerable time reviewing applications and organizing recruitment activities.

THE CHALLENGE

Build CampusLink, a full-stack placement platform connecting students, recruiters and placement administrators. Support job postings, eligibility checks, resume-to-job matching, applications, shortlisting and interview scheduling, with separate role-based dashboards and explainable AI-assisted recommendations.

BASIC FUNCTIONALITIES

- Role-Based Access: Provide separate interfaces and permissions for students, recruiters and placement administrators.
- Student Profiles: Allow students to maintain their educational details, skills, projects, experience and resume.
- Job Management: Allow recruiters to create, edit, publish and close job openings with descriptions, required skills and eligibility criteria.
- Eligibility Engine: Evaluate job-specific requirements such as branch, CGPA, graduation year and backlogs against student profiles.
- Job Discovery: Allow students to search and filter opportunities by role, location, company and eligibility.
- Application Workflow: Allow eligible students to apply for jobs and track statuses such as Applied, Shortlisted, Interview Scheduled, Selected or Rejected.
- Recruiter Dashboard: Allow recruiters to review applications, filter candidates, shortlist applicants and update application statuses.
- Placement Administration: Allow authorized placement administrators to manage student records and monitor active recruitment drives.

ADVANCED FUNCTIONALITIES

- Provide interview date, time and status information for candidates and recruiters.
- Extract skills and qualifications from uploaded PDF resumes.
- Match equivalent skills using TF-IDF, embeddings or semantic similarity.
- Calculate an explainable resume-to-job matching score.
- Identify skill gaps and recommend relevant learning resources.
- Provide notifications when application statuses change or interviews are scheduled.
- Add calendar views for upcoming interviews and recruitment drives.
- Generate analytics showing in-demand skills, application counts and placement outcomes.
- Export candidate lists and recruitment reports.
- Introduce audit logs for important recruiter and administrator actions.

BROWNIE POINTS

- Demonstrate the complete workflow: recruiter publishes a job, student applies, recruiter shortlists and the application status updates.
- Enforce eligibility requirements consistently on both the backend and the user interface.
- Prevent unauthorized users from viewing or modifying restricted student and recruitment data.
- Compare keyword-based matching with semantic matching on sample resumes that use different wording for similar skills.
- Validate uploaded resumes and handle missing, malformed or unsupported files.
- Implement pagination, sorting and server-side filtering for candidate lists.
- Provide clear explanations for matching scores and eligibility failures.
- Include realistic sample student profiles, resumes and job postings for a reliable demonstration.

PS 04

ContradictIQ

MEDIUM

Financial Intelligence & Evidence Investigation Platform

Financial researchers work with reports, spreadsheets and news articles that may contain inconsistent figures, conflicting claims or misleading comparisons. Manually verifying these claims is time-consuming and requires a reliable way to organize documents, trace evidence and record conclusions.

THE CHALLENGE

Build ContradictIQ, a full-stack investigation platform that enables users to upload financial documents, verify claims using deterministic calculations, compare supporting and opposing evidence, and maintain traceable investigation histories. AI may assist with claim extraction and analysis, but conclusions must be linked to source documents and verified calculations.

BASIC FUNCTIONALITIES

- Investigation Management: Create, view, edit and archive financial investigations with titles, descriptions and statuses.
- Document Upload: Upload supported PDF and CSV files, validate file types and sizes, and maintain document metadata.
- Document Workspace: Display uploaded documents and extracted information in an organized interface.
- Claim Management: Allow users to create or extract financial claims containing a metric, value, reporting period and source reference.
- Financial Verification: Recalculate basic metrics such as revenue growth, profit margins and expense ratios using deterministic code.
- Evidence Comparison: Compare claims against reported figures and classify findings as Consistent, Conflicting or Inconclusive.
- Evidence Linking: Link each finding to its source document, relevant passage or table, reporting period and verification calculation.
- Investigation Dashboard: Display findings, unresolved claims, document counts and investigation status.
- Store investigation metadata, claims, verification results and document references in a database.

ADVANCED FUNCTIONALITIES

- Allow users to revisit saved investigations and inspect earlier findings.
- AI-assisted extraction of claims, financial metrics and reporting periods from documents.
- Highlight supporting passages in a document viewer.
- Compare multiple documents and reporting periods side by side.
- Add comments, reviewer assignments and collaborative investigation workflows.
- Maintain version history when claims or conclusions are modified.
- Generate a downloadable investigation report with evidence references.
- Add an evidence graph connecting claims, metrics, source documents and conclusions.
- Use multiple AI agents for evidence retrieval, numerical verification and independent criticism.
- Introduce background document-processing jobs with visible processing status.

BROWNIE POINTS

- Detect differences between quarterly and annual figures without incorrectly treating them as direct contradictions.
- Handle missing, malformed or inconsistent financial data gracefully.
- Demonstrate an example in which an initial claim is disproved by supporting documents.
- Provide traceable calculations rather than relying only on AI-generated explanations.
- Prevent strong conclusions when evidence is incomplete or incomparable.
- Implement access control for private investigations and uploaded documents.
- Measure the correctness of contradiction detection and the false-positive rate on a labelled sample dataset.
- Allow judges to inspect the complete history of an investigation.

PS 05

Strategy Arena

MEDIUM

Interactive Business Strategy Simulation & Decision Platform

Businesses must choose strategies involving pricing, marketing, operating costs and customer retention. Different proposals may prioritize different outcomes, making it difficult to determine which strategy performs best under changing market conditions and resource constraints.

THE CHALLENGE

Build Strategy Arena, a full-stack simulation platform where users create business scenarios, compare competing strategies and evaluate their outcomes through an interactive dashboard. Use a shared deterministic simulation engine to calculate results, enforce constraints, save simulation runs and demonstrate how changing assumptions affects strategy rankings. AI agents may propose or critique strategies as an optional enhancement.

BASIC FUNCTIONALITIES

- Project Management: Create and manage business strategy projects with saved objectives and assumptions.
- Scenario Builder: Configure business parameters such as prices, customer demand, operating costs, marketing expenditure and available capacity.
- Strategy Management: Create, edit and save multiple competing strategies for the same business scenario.
- Simulation Engine: Calculate projected revenue, expenses, profit and other relevant outcomes using a shared deterministic model.
- Scenario Comparison: Compare strategies side by side using charts, tables and key performance indicators.
- Constraint Validation: Identify strategies that violate configured budget, capacity or other business constraints.
- Sensitivity Analysis: Allow users to change assumptions and observe how the results and strategy rankings change.
- Simulation History: Save each simulation run with its input assumptions, strategy version and calculated outcomes.
- Display rankings, trade-offs, constraints and the reasons behind the preferred strategy.

ADVANCED FUNCTIONALITIES

- Specialist AI agents for pricing, growth, operations and financial risk.
- An adversarial reviewer that identifies weaknesses in each proposal.
- Iterative negotiation where agents revise strategies based on critiques.
- Multi-objective optimization balancing profit, cash flow, customer retention and risk.
- Scenario templates for rising costs, falling demand and competitor price changes.
- Team collaboration with comments and shared projects.
- Exportable reports summarizing assumptions, strategy comparisons and recommendations.
- Background simulation jobs for computationally expensive scenarios.
- A visual history showing how a strategy changes between revisions.

BROWNIE POINTS

- Demonstrate a complete workflow from creating a scenario to comparing and saving strategies.
- Show a case where the highest-revenue strategy is not the most profitable or feasible.
- Introduce new evidence and explain why the preferred strategy changes.
- Allow users to reproduce a previous simulation using its saved assumptions.
- Show the effect of changing business priorities on strategy rankings.
- Ensure that all proposals use the same simulation rules and constraints.
- Provide meaningful error messages for invalid scenarios.
- Include a baseline strategy and compare its outcomes against the proposed strategies.

PS 06

ConsultOS

HARD

Autonomous Business Investigation & Workflow Platform

Businesses often investigate complex problems using spreadsheets, reports and internal documents. Fixed workflows may miss important evidence, while basic chatbots can produce unverified explanations. A reliable solution must coordinate analytical tasks, track dependencies and revise its investigation when new evidence appears.

THE CHALLENGE

Build ConsultOS, a full-stack investigation platform that converts business questions into structured workflows, executes approved analytical tools and produces evidence-backed reports. The system should provide visual task dependencies, persistent execution state, live progress updates and controlled replanning, with AI assisting investigation planning and hypothesis generation.

BASIC FUNCTIONALITIES

- Investigation Workspace: Allow users to create, save, reopen and manage business investigations.
- Question and Data Input: Accept a business question and supported CSV, PDF or other predefined input formats.
- Hypothesis Generation: Generate or manually define possible explanations for the observed business problem.
- Investigation Planner: Convert hypotheses into tasks with dependencies, expected outputs and execution statuses.
- Workflow Visualization: Display tasks and their relationships through a visual graph or structured workflow interface.
- Tool Execution: Execute a controlled set of analytical tools, such as calculations, CSV queries, document retrieval and basic statistical analysis.
- Execution Dashboard: Display task statuses, progress, execution logs, errors and unresolved questions.
- Evidence Management: Store findings and link them to the calculations, data records or source documents that support them.
- Final Report: Generate a structured report containing findings, alternative explanations, recommendations and unresolved questions.

ADVANCED FUNCTIONALITIES

- Dynamic replanning when an assumption fails or new evidence contradicts an earlier conclusion.
- Specialist agents for financial analysis, operations, document research and independent review.
- A dependency graph linking hypotheses, tasks, evidence and conclusions.
- Live progress streaming using Server-Sent Events (SSE) or WebSockets.
- Retry policies, timeouts and recovery from failed tasks.
- Configurable execution limits for tool calls, runtime and computational cost.
- Background execution for long-running investigations.
- A replay interface showing how the investigation evolved over time.
- Comparison between a predefined workflow and an adaptive investigation.
- Collaborative workspaces with reviewer comments and investigation sharing.

BROWNIE POINTS

- Provide a complete workflow from a business question to a final evidence-backed report.
- Demonstrate an investigation that changes direction after new evidence is introduced.
- Recover from a failed task without losing previously completed results.
- Enforce dependency ordering so tasks execute only when their prerequisites are satisfied.
- Make completed investigations reproducible and inspectable.
- Track execution time, failed attempts, tool calls and other relevant resource usage.
- Measure conclusion accuracy, evidence coverage, task efficiency and unsupported-claim rate.
- Ensure that the system uses only approved tools and respects configured execution limits.

PS 07

SignalForge

HARD

Quantitative Research & Experiment Management Terminal

Quantitative researchers use historical market data to develop trading signals, train models and evaluate hypothetical strategies. Disconnected scripts and spreadsheets make it difficult to reproduce experiments, compare performance fairly and identify models that fail on unseen data.

THE CHALLENGE

Build SignalForge, a full-stack research platform where users upload historical market data, configure experiments, evaluate predictive signals and compare hypothetical trading performance. Support data validation, chronological model evaluation, backtesting with transaction costs and reproducible experiment histories, while preventing data leakage and look-ahead bias.

BASIC FUNCTIONALITIES

- Research Workspace: Create and manage research projects and experiments.
- Dataset Management: Upload historical price and volume data through CSV files and validate the required columns, timestamps and missing values.
- Data Preview: Display dataset summaries, missing-value statistics and price or volume charts.
- Feature Engineering: Calculate basic indicators such as returns, moving averages, momentum and volatility.
- Experiment Configuration: Allow users to select features, model settings, target definitions and evaluation periods.
- Model Evaluation: Train and compare at least one baseline predictive model against a simple benchmark.
- Chronological Validation: Separate training and testing periods by time to reduce data leakage.
- Backtesting Engine: Convert predictions into hypothetical trading signals and calculate portfolio returns using configurable transaction costs.
- Performance Dashboard: Display predictive metrics alongside cumulative returns, drawdown and other relevant performance indicators.
- Experiment History: Save experiment configurations, dataset references, model settings and results for later comparison.

ADVANCED FUNCTIONALITIES

- Compare multiple models such as logistic regression, random forests and gradient boosting.
- Perform rolling-window evaluation across multiple time periods.
- Test whether signals generalize to unseen stocks or assets.
- Compare performance across different market regimes.
- Add allocation constraints and combine multiple signals into a portfolio.
- Introduce background jobs for training and backtesting.
- Provide live progress updates for running experiments.
- Add experiment versioning and reproducibility checks.
- Generate downloadable research reports with assumptions and evaluation details.
- Build a leaderboard that rewards robust performance rather than raw training accuracy.

BROWNIE POINTS

- Demonstrate a signal that performs well on training data but fails on unseen data, and explain why.
- Prevent future information from leaking into historical predictions.
- Include realistic transaction costs in backtesting.
- Compare a predictive model against a simple benchmark.
- Allow users to reproduce saved experiments from their recorded configurations.
- Compare results across multiple stocks or time periods.
- Handle malformed datasets, missing timestamps and invalid values gracefully.
- Distinguish model accuracy, strategy returns and risk-adjusted performance in the dashboard.
- Provide a report describing assumptions, evaluation methodology and limitations.